

Fluid Flow in Pipes

Pre-Lab Plan

Unit Operations Lab # 1

09/09/2025

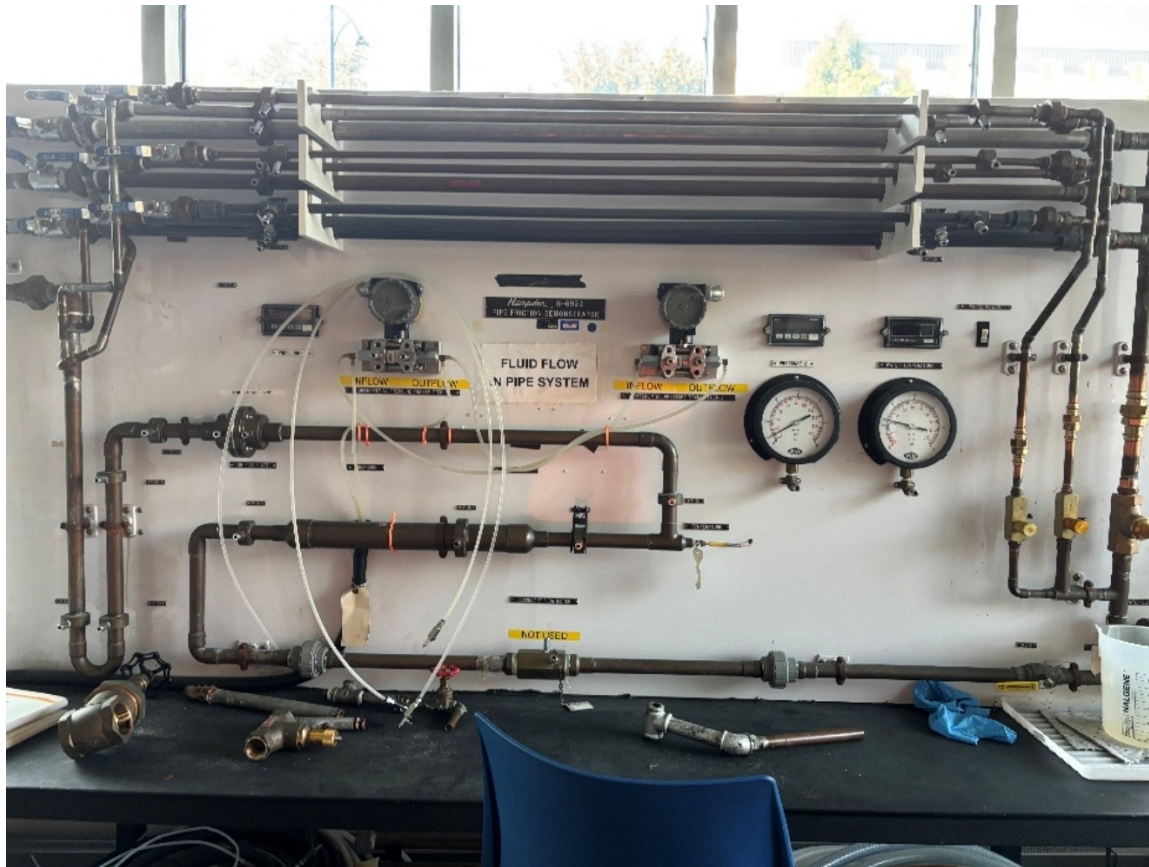
Group #3, Tuesday 9 AM

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1. Experimental Objective

The purpose of the Fluid-Flow in Pipes Lab is to demonstrate how pipe systems are directly involved in transporting materials. This explains the need for different-sized pipes for certain types of flow into reactors, separation processes, and storage facilities. The importance of this lab and its application is to demonstrate how detrimental the engineering design of the pipes is for a plant to commit certain actions. It also shows the importance of energy loss coefficients of fluid flow in a pipe, such as pressure drop, friction factor, loss factor, and flow coefficient. Pressure drop is a necessary calculation to be performed to ensure proper equipment is placed in the correct position of the process, such as a pump. The unit ops lab allows the performance of this application to test certain theories and correlations tied to fluid flow in a pipe. The types of data that are captured in this lab are pressure drop, flow rate, temperature, and tube measurements.

2.1 Apparatus

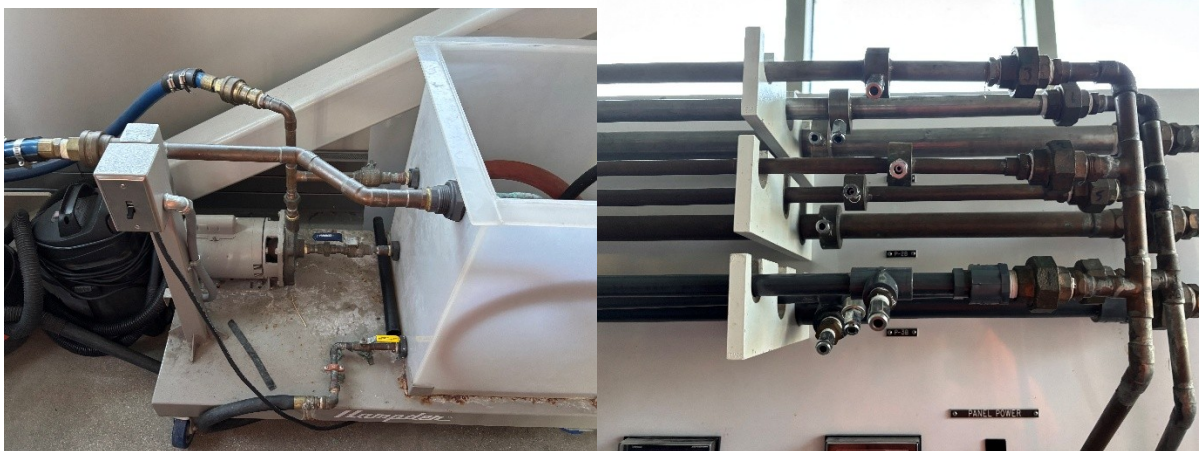


The fluid flow system consists of a Hampden H-6920 Pipe Friction Demonstrator; a bench mounted panel upon which is located the flow circuits and instrumentation, as well as a separate pump, reservoir tank and rotameter to control the flow of water into and out of the system. The

flow circuit is comprised of one main path with nine parallel branches at the top.



Rotometer and water reservoir tank



Water pump

nine parallel branches of tubing

The pipe lengths used for the pipe friction head loss experiments are the nine parallel branches at the top consisting of copper tubing, plastic pipe, and stainless steel pipes.



The quickconnect water plugs and the flow control valves are in the lower right corner.



The shut-off valves are in the upper left corner

2.2 Materials and supplies

Material needed	Use
Calipers	Measure the diameter of the pipe
Water	Fluid-Flow
Tape Measure	Length of the pipe

2.3 Experimental Plan

Procedure

Rotameter Calibration:

1. Fill the tank with water while making sure the discharge hose is closed
2. After the tank is full, turn the pump on, open the discharge hose and use it to pour water into a bucket, all the while keeping track of the time. Keep note of the amount of water collected and how long it took to collect that amount
3. Graph it in Excel (L y-axis and s x-axis) and get the slope of the line. Also, check the R^2 value and make sure it's close to 1. The slope is in LPS, so convert it to LPM.
Afterwards, multiply the LPM with 0.8455 and that is the new conversion from SCFM to LPM

Orifice Meter:

1. Connect DPT across orifice taps
2. Take 8-10 flow setpoints for the selected pipe, while making sure Re is reasonable
3. Record flowrate, differential pressure across orifice plate, and temperature

4. Solve for C_f using
$$C_f = \frac{Q}{A_0 \sqrt{\frac{2 \Delta P}{\rho}}}$$
 for all trials

5. Make a note of patterns

Pipe Friction Head Loss:

1. Choose 3 branches covering S/M/L and at least 2 materials
2. Connect plastic tube to the upstream/downstream taps of the chosen pipe section
3. For each branch and 8-10 points, measure differential pressure drops over the straight length of piping, Q, and T

4. Find; $v = \frac{4Q}{\pi D^2}$, Re, and $f = \frac{\Delta P}{\left(\frac{L}{D}\right) \left(\frac{\rho v^2}{2}\right)}$. Compare with $f = \frac{64}{Re}$

Pipe Fitting Head Loss:

1. Look at the different fittings (bends, elbows, expansion, and contraction) and decide the order of testing
2. For the fitting being tested, connect DPT
3. Use 6-8 setpoints and record differential pressure, ΔP , Q, and T
4. Calculate: $K = \frac{fL}{D}$, $h_L = \frac{K V^2}{2}$

Variables

- Flow rate, pipe size and material, and fitting type are independent variables
- Pressures and temperatures are dependent

Target Ranged

- The Re for laminar should be less than 1800
- The Re for turbulent should be more than 4000

Calibration Method

- Rotometer Calibration (mentioned earlier)

Project Safety Evaluation

	<i>Yes/No</i>	
Potential electrical hazards?	yes	If yes, identify conditions: water tank near electrical components
Potential mechanical hazards?	yes	If yes, identify conditions: air in pump, pressure build up in pipes
Potential pressure hazards?	no	If yes, identify conditions:
Potential temperature hazards?	no	If yes, identify conditions:
Hazardous/toxic chemicals involved?	no	If yes, hazards involved:
Gloves required?	yes	If yes, specify type:
MSDS reviewed by all team members?		List MSDS sheets reviewed:
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	32.2	
Pressure (psia)	25	
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>

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I have read relevant background material for the Unit Operations Laboratory entitled:
“_____” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

<u>Anna Cai</u>	signature of team member	<u>09/</u> date
<u>Hussein Rezk</u>	signature of team member	<u> </u> date
<u>Aaron Casas</u>	signature of team member	<u> </u> date
<u>Adil Feratovic</u>	signature of team member	<u> </u> date